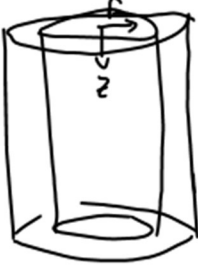


2. By making a mass balance over a volume element $(\Delta r)(r\Delta\theta)(\Delta z)$ derive the equation of continuity in cylindrical coordinates.

$$\frac{\partial \rho}{\partial t} \Delta r r \Delta \theta \Delta z = -(\text{mass rate out})$$

$$\Delta V = \Delta r r \Delta \theta \Delta z$$



$$\rho v_r \cdot r \Delta \theta \Delta z \Big|_{r=r+\Delta r} - \rho v_r \cdot r \Delta \theta \Delta z \Big|_{r=r} = \frac{\partial}{\partial r} (\rho v_r r) \Delta \theta \Delta z \Delta r$$

$$\rho v_\theta \Delta r \Delta z \Big|_{\theta=\theta+\Delta\theta} - \rho v_\theta \Delta r \Delta z \Big|_{\theta=\theta} = \frac{\partial}{\partial \theta} (\rho v_\theta) \Delta r \Delta z \Delta \theta$$

$$\rho v_z r \Delta r \Delta \theta \Big|_{z=z+\Delta z} - \rho v_z r \Delta r \Delta \theta \Big|_{z=z} = \frac{\partial}{\partial z} (v_z \rho) \Delta r \Delta \theta \Delta z r$$



Mass Balance:

$$\frac{\partial \rho}{\partial t} \Delta r r \Delta \theta \Delta z = - \frac{\partial}{\partial r} (\rho v_r r) \Delta \theta \Delta z \Delta r - \frac{\partial}{\partial \theta} (\rho v_\theta) \Delta r \Delta z \Delta \theta - \frac{\partial}{\partial z} (v_z \rho) \Delta r \Delta \theta \Delta z r$$

$$r \frac{\partial \rho}{\partial t} = - \frac{\partial}{\partial r} (\rho v_r r) - \frac{\partial}{\partial \theta} (\rho v_\theta) - \frac{\partial}{\partial z} (v_z \rho) \cdot r \quad | \cdot \frac{1}{r}$$

$$\Rightarrow \frac{\partial \rho}{\partial t} = - \frac{1}{r} \frac{\partial}{\partial r} (\rho v_r r) - \frac{1}{r} \frac{\partial}{\partial \theta} (\rho v_\theta) - \frac{\partial}{\partial z} (v_z \rho)$$

$$\Rightarrow \boxed{\frac{\partial \rho}{\partial t} + \frac{1}{r} \frac{\partial}{\partial r} (\rho v_r r) + \frac{1}{r} \frac{\partial}{\partial \theta} (\rho v_\theta) + \frac{\partial}{\partial z} (v_z \rho) = 0}$$

3A.2 Friction loss in bearings.¹ Each of two screws on a large motor-ship is driven by a 4000-hp engine. The shaft that connects the motor and the screw is 16 in. in diameter

and rests in a series of sleeve bearings that give a 0.005 in. clearance. The shaft rotates at 50 rpm, the lubricant has a viscosity of 5000 cp, and there are 20 bearings, each 1 ft in length. Estimate the fraction of engine power expended in rotating the shafts in their bearings. Neglect the effect of the eccentricity.

$$\mu = 5000 \text{ cp}$$

$$v_\theta(KR) = \omega KR$$

$$v_\theta(R) = 0$$



Cylindrical Coordinates Navier-Stokes - equation of motion, constant μ & ρ

r-component $-\rho \frac{v_\theta^2}{r} = -\frac{\partial p}{\partial r}$

θ -component $0 = \mu \frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r} (r v_\theta) \right)$

z -component $0 = -\frac{\partial p}{\partial z} - \rho g_z$

$\int 0^r = \int \frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r} (r v_\theta) \right) dr$

$$C_1 = \frac{1}{r} \frac{d}{dr} (r v_\theta) \Rightarrow \int C_1 r^{dv} = \int \frac{d}{dr} (r v_\theta) dr \Rightarrow \frac{C_1 r^2}{2} + C_2 = r v_\theta \Rightarrow v_\theta = \frac{C_1 r}{2} + \frac{C_2}{r}$$

$$v_\theta(R) = 0 \quad v_\theta(R) = 0 = \frac{C_1 R}{2} + \frac{C_2}{R} \Rightarrow C_2 = -\frac{C_1 R^2}{2}$$

$$v_\theta(KR) = \omega KR \quad v_\theta(KR) = \omega KR = \frac{C_1 KR}{2} + \frac{C_2}{KR} \Rightarrow \omega KR = \frac{C_1 KR}{2} - \frac{C_1 R^2}{2 KR} = \frac{C_1}{2} \frac{K^2 R^2 - R^2}{KR} = \omega KR$$

$$\Rightarrow 2 \omega K^2 R^2 = C_1 (K^2 R^2 - R^2) \Rightarrow C_1 = \frac{2 \omega K^2}{K^2 - 1}, \quad C_2 = -\frac{\omega K^2 R^2}{K^2 - 1}$$

$$\Rightarrow v_{\theta}(r) = \frac{\Omega K^2 r}{K^2 - 1} - \frac{\Omega K^2 R^2}{r(K^2 - 1)} = \frac{\Omega K^2}{K^2 - 1} \left(\frac{r^2 - R^2}{r} \right)$$

torque on inside: $-T_{r\theta}|_{r=KR} \cdot A = -T_{r\theta}|_{r=KR} \cdot 2\pi KR \cdot L = \mu r \frac{d}{dr} \left(\frac{v_{\theta}}{r} \right) \Big|_{r=KR} \cdot 2\pi KR L \cdot KR$

$$T_{r\theta} = -\mu r \frac{d}{dr} \left(\frac{v_{\theta}}{r} \right)$$

$$T_{r\theta}|_{r=KR} = -\mu r \frac{d}{dr} \left(\frac{\Omega K^2}{K^2 - 1} \left(\frac{r^2 - R^2}{r^2} \right) \right) \Big|_{r=KR} = -\mu \frac{K^2 R \Omega}{K^2 - 1} \frac{d}{dr} \left(1 - \frac{R^2}{r^2} \right) \Big|_{r=KR} = -\mu \frac{K^2 R \Omega}{K^2 - 1} \left(0 + \frac{2R^2}{r^3} \right) = -\frac{4\mu \Omega R^2}{K^2 - 1}$$

$$\rightarrow - \left(-\frac{4\mu \Omega R^2}{K^2 - 1} \right) \cdot 2\pi KR L = \frac{4\pi \mu \Omega K^2 R^2 L}{K^2 - 1}$$

thus torque required to maintain motion = $\frac{4\pi \mu K^2 R^2 L}{1 - K^2} \Omega$

$$P = \text{torque} \cdot \Omega \Rightarrow P = \frac{4\pi \mu K^2 R^2 L}{1 - K^2} \Omega^2$$

$$\Omega = 50 \frac{\text{rev}}{\text{min}} \times \frac{2\pi \text{ rad}}{1 \text{ rev}} \times \frac{1 \text{ min}}{60 \text{ s}} = 5.23599 \frac{\text{rad}}{\text{s}}$$

$$L = 20 \text{ bearings} \times \frac{1 \text{ ft}}{\text{bearing}} \times \frac{0.3048 \text{ m}}{1 \text{ ft}} = 6.096 \text{ m}$$

$$R = 8.005 \text{ in} \times \frac{0.0254 \text{ m}}{1 \text{ in}} = 0.20333 \text{ m}$$

$$\mu = 5000 \text{ cP} \times \frac{10^{-3} \text{ Pa}\cdot\text{s}}{1 \text{ cP}} = 5 \text{ Pa}\cdot\text{s}$$

$$K = \frac{rR}{R} = \frac{8 \text{ in}}{8.005 \text{ in}} = 0.9993753$$

$$P = \frac{4(5 \text{ Pa}\cdot\text{s}) \cdot \pi \cdot (0.9993753)^2 \cdot (0.20333 \text{ m})^2 \cdot 6.096 \text{ m}}{1 - (0.9993753)^2} \cdot (5.23599 \frac{\text{rad}}{\text{s}})^2$$

$$P \approx 347150 \text{ W} = 347.15 \text{ kW}$$

$$347.15 \text{ kW} \times \frac{1.341 \text{ hp}}{1 \text{ kW}} = 465.53 \text{ hp}$$

fraction of power lost to friction:

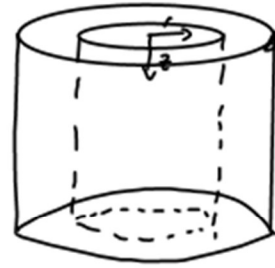
$$\frac{465.53 \text{ hp}}{4000 \text{ hp}} = 0.116$$

3B.1 Flow between coaxial cylinders and concentric spheres.

(a) The space between two coaxial cylinders is filled with an incompressible fluid at constant temperature. The radii of the inner and outer wetted surfaces are κR and R , respectively. The angular velocities of rotation of the inner and outer cylinders are Ω_i and Ω_o . Determine the velocity distribution in the fluid and the torques on the two cylinders needed to maintain the motion.

(b) Repeat part (a) for two concentric spheres.

a.)



$$\theta\text{-Component: } 0 = \mu \left[\frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r} (r v_\theta) \right) \right]$$

$$\int 0 dr = \int \frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r} (r v_\theta) \right) dr$$

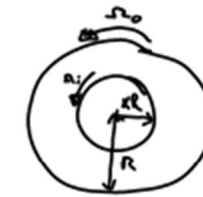
$$C_1 = \frac{1}{r} \frac{\partial}{\partial r} (r v_\theta) \Rightarrow \int r C_1 dr = \int \frac{\partial}{\partial r} (r v_\theta) dr \Rightarrow \frac{r^2 C_1}{2} + C_2 = r v_\theta$$

$$v_\theta(\kappa R) = \Omega_i \kappa R = \frac{C_1 \kappa R}{2} + \frac{C_2}{\kappa R} \Rightarrow \Omega_i \kappa R = \frac{C_1 \kappa R}{2} + \frac{1}{\kappa R} (\Omega_o R^2 - \frac{C_1 R^2}{2})$$

$$v_\theta(R) = \Omega_o R = \frac{C_1 R}{2} + \frac{C_2}{R} \Rightarrow C_2 = \Omega_o R^2 - \frac{C_1 R^2}{2} = R^2 \left(\Omega_o - \frac{\Omega_i \kappa^2 - \Omega_o}{\kappa^2 - 1} \right)$$

$$C_1 = R^2 \left(\frac{\kappa^2 \Omega_o - \Omega_i - \kappa^2 \Omega_o}{\kappa^2 - 1} \right)$$

$$C_2 = R^2 \kappa^2 \left(\frac{\Omega_o - \Omega_i}{\kappa^2 - 1} \right)$$



$$\Omega_i \kappa R - \Omega_o R = C_1 \left(\frac{R(\kappa^2 - 1)}{2\kappa} \right)$$

$$C_1 = \frac{2(\Omega_i \kappa^2 - \Omega_o)}{(\kappa^2 - 1)}$$

$$v_\theta(r) = \frac{r(\Omega_i \kappa^2 - \Omega_o)}{\kappa^2 - 1} + \frac{R^2 \kappa^2}{r} \left(\frac{\Omega_o - \Omega_i}{\kappa^2 - 1} \right)$$

Inner cylinder:

$$\tau_{\theta r} = -\mu \left(r \frac{\partial}{\partial r} \left(\frac{v_\theta}{r} \right) + \frac{1}{r} \frac{\partial v_\theta}{\partial \theta} \right) \Rightarrow \tau_{\theta r} = \mu \left(2r \frac{R^2 \kappa^2}{r^2} \left(\frac{\Omega_o - \Omega_i}{\kappa^2 - 1} \right) \right) = \frac{2\mu R^2 \kappa^2}{r^2} \left(\frac{\Omega_o - \Omega_i}{\kappa^2 - 1} \right) = \tau_{\theta r}$$

$$\frac{\partial}{\partial r} \left(\frac{v_\theta}{r} \right) = 0 - 2 \frac{R^2 \kappa^2}{r^3} \left(\frac{\Omega_o - \Omega_i}{\kappa^2 - 1} \right)$$

$$-\tau_{\theta r}|_{r=\kappa R} \cdot 2\pi \kappa R \cdot \kappa R L = -2\mu \left(\frac{\Omega_o - \Omega_i}{\kappa^2 - 1} \right) 2\pi \kappa^2 R^2 L = -4\pi \mu \kappa^2 R^2 \left(\frac{\Omega_o - \Omega_i}{\kappa^2 - 1} \right) L$$

$\therefore$ torque needed to be applied on the inner cylinder to maintain motion is $4\pi \mu \kappa^2 R^2 L \left(\frac{\Omega_o - \Omega_i}{\kappa^2 - 1} \right)$

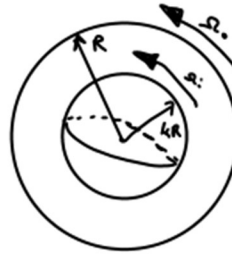
Outer cylinder:

$$-\tau_{\theta r}|_{r=R} \cdot 2\pi R \cdot R L = -2\mu \kappa^2 \left(\frac{\Omega_o - \Omega_i}{\kappa^2 - 1} \right) \cdot 2\pi R^2 L = -4\pi \mu \kappa^2 R^2 L \left(\frac{\Omega_o - \Omega_i}{\kappa^2 - 1} \right)$$

$\therefore$ torque needed to be applied on the outer cylinder to maintain motion is $4\pi \mu \kappa^2 R^2 L \left(\frac{\Omega_o - \Omega_i}{\kappa^2 - 1} \right)$

b.)

$$\gamma_{\theta r} = -M \left(r \frac{\partial}{\partial r} \left(\frac{v_{\theta}}{r} \right) \right)$$



$$\phi \text{ component: } 0 = M \left[\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial v_{\theta}}{\partial r} \right) + \frac{1}{r^2} \frac{\partial}{\partial \theta} \left(\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} (v_{\theta} \sin \theta) \right) \right]$$

$$0 = \frac{\partial}{\partial r} \left(r^2 \frac{\partial v_{\theta}}{\partial r} \right) + \frac{\partial}{\partial \theta} \left(\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} (v_{\theta} \sin \theta) \right), \quad v_{\theta}(r, \theta) = f(r) \sin \theta$$

$$0 = \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} (f(r) \sin \theta) \right) + \frac{\partial}{\partial \theta} \left(\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} (f(r) \sin^2 \theta) \right)$$

$$\frac{\partial}{\partial r} (r^2 f'(r) \sin \theta) + \frac{\partial}{\partial \theta} \left(\frac{1}{\sin \theta} \cdot 2 \sin \theta \cos \theta f(r) \right)$$

$$2r f'(r) \sin \theta + r^2 f''(r) \sin \theta - 2 \cos \theta f(r) = 0$$

$$\Rightarrow r^2 f''(r) + 2r f'(r) - 2f(r) = 0$$

guess $f(r) = r^n$ $\Rightarrow r^2 \cdot n(n-1)r^{n-2} + 2r \cdot n r^{n-1} - 2r^n = 0$

$$f'(r) = n r^{n-1} \Rightarrow n^2 - n + 2n - 2 = 0$$

$$f''(r) = (n-1)r^{n-2} \Rightarrow n^2 + n - 2 = 0 \Rightarrow (n+2)(n-1) = 0, \quad n = 2, 1$$

B.C.

$$v_{\theta}(R, \theta) = \Omega_0 R \sin \theta \Rightarrow f(R) = \Omega_0 R \quad f(r) = C_1 r + \frac{C_2}{r^2}$$

$$v_{\theta}(R, \theta) = \Omega_0 R \sin \theta \Rightarrow f(R) = \Omega_0 R$$

$$f(R) = \Omega_0 R = C_1 R + \frac{C_2}{R^2}$$

$$f(R) = \Omega_0 R = C_1 R + \frac{C_2}{R^2} \Rightarrow C_1 = \Omega_0 - \frac{C_2}{R^3}$$

$$(\Omega_0 - C_1) R^3 = C_2 = (\Omega_0 - \Omega_0 + \frac{C_2}{R^3}) R^3 = C_2$$

$$R^3 (\Omega_0 - \Omega_0) = C_2 - C_2 R^3$$

$$= C_2 (1 - R^3)$$

$$\frac{R^3 (\Omega_0 - \Omega_0)}{1 - R^3} = \frac{C_2}{1 - R^3}$$

$$\Omega_0 - \frac{R^3 (\Omega_0 - \Omega_0)}{1 - R^3} = C_1$$

$$C_1 = \frac{\Omega_0 - \frac{R^3 (\Omega_0 - \Omega_0)}{1 - R^3}}{1 - R^3}$$

$$\underline{C_1 = \frac{\Omega_0 - R^3 \Omega_0}{1 - R^3}}$$

$$f(r) = \frac{\Omega_0 - R^3 \Omega_0}{1 - R^3} r + \frac{R^3 R^3 (\Omega_0 - \Omega_0)}{(1 - R^3) r^2}$$

$$v_{\theta}(r, \theta) = f(r) \sin \theta$$

$$\Rightarrow v_{\theta}(r, \theta) = \frac{1}{1 - R^3} \left((\Omega_0 - R^3 \Omega_0) r + R^3 R^3 (\Omega_0 - \Omega_0) \left(\frac{1}{r^2} \right) \right) \sin \theta$$

Inner sphere:

$$\begin{aligned} \vec{T}_i &= \iint_S -\tau_{\theta\phi} \big|_{r=R} \cdot \vec{R} \sin\theta dS = \int_0^{2\pi} \int_0^\pi -\tau_{\theta\phi} \big|_{r=R} \cdot \vec{R} \sin\theta \cdot R^2 \sin\theta d\theta d\phi = -K^2 R^3 \int_0^{2\pi} \int_0^\pi \mu r \frac{\partial}{\partial r} \left(\frac{\nabla\theta}{r} \right) \bigg|_{r=R} \sin^2\theta d\theta d\phi \\ \Rightarrow K^2 R^3 \int_0^{2\pi} \int_0^\pi r \cdot \left(-3 \frac{K^2 R^3 (\alpha; -\alpha_0)}{(1-K^2) r^3} \right) \bigg|_{r=R} \sin^2\theta d\theta d\phi &= K^2 R^3 \mu \int_0^{2\pi} \int_0^\pi \left(-3 \frac{\alpha; -\alpha_0}{1-K^2} \right) \cdot \sin^2\theta d\theta d\phi \\ \Rightarrow \frac{-K^2 R^3 \mu 6\pi (\alpha; -\alpha_0)}{1-K^2} \cdot \underbrace{\int_0^\pi \sin^2\theta d\theta}_{=\frac{\pi}{2}} &= -\frac{8 K^2 R^3 \mu \pi (\alpha; -\alpha_0)}{1-K^2} = \vec{T}_i \end{aligned}$$

∴ torque needed to be applied on the inner sphere to maintain motion is $\frac{8 K^2 R^3 \mu \pi (\alpha; -\alpha_0)}{1-K^2}$

Outer sphere:

$$\begin{aligned} \vec{T}_o &= \iint_S -\tau_{\theta\phi} \big|_{r=R} \cdot \vec{R} \sin\theta dS = \int_0^{2\pi} \int_0^\pi \mu r \frac{\partial}{\partial r} \left(\frac{\nabla\theta}{r} \right) \bigg|_{r=R} \cdot \vec{R} \sin\theta R^2 \sin\theta d\theta d\phi = 2\pi \mu \int_0^\pi \left(-3 \frac{K^2 R^3 (\alpha; -\alpha_0)}{(1-K^2) r^3} \right) \bigg|_{r=R} R^3 \sin^3\theta d\theta \\ \Rightarrow \frac{-3 K^2 R^3 (\alpha; -\alpha_0)}{1-K^2} \cdot 2\pi \mu \int_0^\pi \sin^3\theta d\theta &= -\frac{8 \pi \mu K^2 R^3 (\alpha; -\alpha_0)}{1-K^2} = \vec{T}_o \end{aligned}$$

∴ torque needed to be applied on the outer sphere to maintain motion is $\frac{8 K^2 R^3 \mu \pi (\alpha; -\alpha_0)}{1-K^2}$

3B.4 Creeping flow between two concentric spheres (Fig. 3B.4). A very viscous Newtonian fluid flows in the space between two concentric spheres, as shown in the figure. It is desired to find the rate of flow in the system as a function of the imposed pressure difference. Neglect end effects and postulate that v_θ depends only on r and θ with the other velocity components zero.

(a) Using the equation of continuity, show that $v_\theta \sin \theta = u(r)$, where $u(r)$ is a function of r to be determined.

(b) Write the θ -component of the equation of motion for this system, assuming the flow to be slow enough that the $[\mathbf{v} \cdot \nabla \mathbf{v}]$ term is negligible. Show that this gives

$$0 = -\frac{1}{r} \frac{\partial p}{\partial \theta} + \mu \left[\frac{1}{\sin \theta} \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{du}{dr} \right) \right] \quad (3B.4-1)$$

(c) Separate this into two equations

$$\sin \theta \frac{dp}{d\theta} = B; \quad \frac{\mu}{r} \frac{d}{dr} \left(r^2 \frac{du}{dr} \right) = B \quad (3B.4-2, 3)$$

where B is the separation constant, and solve the two equations to get

$$B = \frac{p_2 - p_1}{2 \ln \cot \frac{\epsilon}{2}} \quad (3B.4-4)$$

$$u(r) = \frac{(p_1 - p_2)R}{4\mu \ln \cot(\epsilon/2)} \left[\left(1 - \frac{r}{R} \right) + \kappa \left(1 - \frac{R}{r} \right) \right] \quad (3B.4-5)$$

where p_1 and p_2 are the values of the modified pressure at $\theta = \epsilon$ and $\theta = \pi - \epsilon$, respectively.

(d) Use the results above to get the mass rate of flow

$$w = \frac{\pi(p_1 - p_2)R^3(1 - \kappa)^3 \rho}{12\mu \ln \cot(\epsilon/2)} \quad (3B.4-6)$$

12.) $v_\theta(r, \theta)$ $\frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (v_\theta \sin \theta) = 0 \Rightarrow \frac{\partial}{\partial \theta} (v_\theta \sin \theta) = 0 \Rightarrow \int \frac{\partial}{\partial \theta} (v_\theta \sin \theta) d\theta = \int d\theta$

B.C:

$$v_\theta(r, \theta) = 0$$

$$v_\theta(R, \theta) = 0$$

$$v_\theta \sin \theta = u(r)$$

11 b.) $\rho \left(\cancel{\frac{\partial v_\theta}{\partial t}} + v_r \frac{\partial v_\theta}{\partial r} + \frac{v_\theta}{r} \frac{\partial v_\theta}{\partial \theta} + \frac{v_\theta}{r \sin \theta} \frac{\partial v_\theta}{\partial \phi} + \frac{v_r v_\theta - v_\theta^2 \cot \theta}{r} \right) = -\frac{1}{r} \frac{\partial p}{\partial \theta} + \mu \left(\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial v_\theta}{\partial r} \right) + \frac{1}{r^2} \frac{\partial}{\partial \theta} \left(\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} (v_\theta \sin \theta) \right) \right)$

$$0 = -\frac{1}{r} \frac{\partial p}{\partial \theta} + \mu \left[\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial v_\theta}{\partial r} \right) \right] + \rho g_\theta$$

$$v_\theta \sin \theta = u(r) \Rightarrow v_\theta = \frac{u(r)}{\sin \theta}, \quad g_\theta = g \sin \theta$$

$$\Rightarrow 0 = -\frac{1}{r} \frac{\partial p}{\partial \theta} + \rho g \sin \theta + \mu \left[\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \left(\frac{u(r)}{\sin \theta} \right) \right) \right]$$

$$0 = -\frac{1}{r} \underbrace{\left(\frac{\partial p}{\partial \theta} - \rho g \sin \theta \right)}_{\frac{\partial p}{\partial \theta}} + \mu \left[\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \left(\frac{u(r)}{\sin \theta} \right) \right) \right]$$

$$0 = -\frac{1}{r} \frac{\partial p}{\partial \theta} + \mu \left[\frac{1}{\sin \theta} \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{du}{dr} \right) \right]$$

11 C.)

$$\theta_1 = \xi, \theta_2 = \pi - \xi$$

$$\frac{1}{r} \frac{\partial \mathcal{P}}{\partial \theta} = \frac{M}{r^2 \sin \theta} \frac{d}{dr} \left(r \frac{d\mathcal{U}}{dr} \right)$$

$$\Rightarrow \sin \theta \frac{\partial \mathcal{P}}{\partial \theta} = \frac{M}{r} \frac{d}{dr} \left(r \frac{d\mathcal{U}}{dr} \right) \quad \sin \theta \frac{\partial \mathcal{P}}{\partial \theta} = B \quad \xrightarrow{\text{Separation constant}} \int_{\mathcal{P}_1}^{\mathcal{P}_2} d\mathcal{P} = \int_{\theta_1}^{\theta_2} \frac{B}{\sin \theta} d\theta$$

$$\Rightarrow \mathcal{P}_2 - \mathcal{P}_1 = B \ln \left| \tan \frac{\theta}{2} \right| \Big|_{\xi}^{\pi - \xi}$$

$$= B \ln \left| \frac{\tan(\frac{\pi - \xi}{2})}{\tan(\frac{\xi}{2})} \right|$$

$$= B \ln \left| \cot^2 \left(\frac{\xi}{2} \right) \right| = 2B \ln \left| \cot \left(\frac{\xi}{2} \right) \right|$$

$$\Rightarrow \frac{\mathcal{P}_2 - \mathcal{P}_1}{2 \ln \left| \cot \left(\frac{\xi}{2} \right) \right|} = B$$

$$\frac{M}{r} \frac{d}{dr} \left(r \frac{d\mathcal{U}}{dr} \right) = B$$

$$\Rightarrow \int \frac{d}{dr} \left(r \frac{d\mathcal{U}}{dr} \right) dr = \int \frac{B r}{M} dr$$

$$\Rightarrow r \frac{d\mathcal{U}}{dr} = \frac{B r^2}{2M} + C_1$$

$$\Rightarrow \int \frac{d\mathcal{U}}{dr} = \frac{B}{2M} + \frac{C_1}{r^2} dr$$

$$\mathcal{U}(r) = \frac{B r}{2M} - \frac{C_1}{r} + C_2$$

$$\mathcal{U}(KR) = 0 = \frac{B KR}{2M} - \frac{C_1}{KR} + C_2$$

$$\mathcal{U}(R) = 0 = \frac{B R}{2M} - \frac{C_1}{R} + C_2$$

$$C_2 = \frac{C_1}{R} - \frac{B R}{2M}$$

$$\Rightarrow \frac{B KR}{2M} - \frac{C_1}{KR} + \frac{C_1}{R} - \frac{B R}{2M} = 0$$

$$\Rightarrow C_1 \left(\frac{1}{R} - \frac{1}{KR} \right) = \frac{B R}{2M} - \frac{B KR}{2M} \Rightarrow C_1 = \frac{B R (1-K)}{2M} \cdot \frac{KR}{K-1} = -\frac{B R^2 K}{2M}$$

$$C_2 = -\frac{B R K}{2M} - \frac{B R}{2M} = -\frac{B R (K+1)}{2M}$$

$$\mathcal{U}(r) = \frac{B r}{2M} + \frac{B R^2 K}{2M r} - \frac{B R (K+1)}{2M} = \frac{B}{2M} \left(r + \frac{R^2 K}{r} - R(K+1) \right)$$

$$= \frac{(\mathcal{P}_2 - \mathcal{P}_1) R}{2 \ln \left| \cot \left(\frac{\xi}{2} \right) \right| \cdot 2M} \left(\frac{r}{R} + \frac{R K}{r} - K - 1 \right) = \frac{(\mathcal{P}_2 - \mathcal{P}_1) R}{4M \ln \cot \left(\frac{\xi}{2} \right)} \left(1 - \frac{r}{R} + K \left(1 - \frac{R}{r} \right) \right)$$

$$\mathcal{U}(r) = \frac{(\mathcal{P}_2 - \mathcal{P}_1) R}{4M \ln \cot \left(\frac{\xi}{2} \right)} \left(\left(1 - \frac{r}{R} \right) + K \left(1 - \frac{R}{r} \right) \right)$$

Mass flow rate:

$$\omega = \rho \int_A \mathcal{V} dA$$

$$dA = r \sin \theta dr d\theta$$

$$= \rho \int_0^{2\pi} \int_{KR}^R \frac{\mathcal{U}(r)}{\sin \theta} r \sin \theta dr d\theta = 2\pi \rho \cdot \frac{(\mathcal{P}_1 - \mathcal{P}_2) R}{4M \ln \cot \left(\frac{\xi}{2} \right)} \int_{KR}^R \left[\left(1 - \frac{r}{R} \right) + K \left(1 - \frac{R}{r} \right) \right] dr$$

$$\int_{KR}^R \left(r - \frac{r^2}{R} + KR - \frac{KR^2}{r} \right) dr$$

$$\left. \frac{r^2}{2} - \frac{r^3}{3R} + \frac{KR^2}{2} - KRr \right|_{KR}^R$$

$$\frac{R^2(1-K)^3}{6}$$

$$\omega = \frac{\rho \pi (\mathcal{P}_1 - \mathcal{P}_2) R}{2M \ln \cot \left(\frac{\xi}{2} \right)} \cdot \frac{R^2(1-K)^3}{6}$$

$$\omega = \frac{\pi R^3 (\mathcal{P}_1 - \mathcal{P}_2) (1-K)^3 \rho}{12M \ln \cot \left(\frac{\xi}{2} \right)}$$

1- Using the continuity equation $\frac{\partial \rho}{\partial t} = -\text{div } \rho \langle v |$, prove:

$$1.) \quad \frac{\partial}{\partial t} \rho \frac{v^2}{2} + \text{div } \rho \frac{v^2}{2} \langle v | = \rho \frac{D}{Dt} \frac{v^2}{2}$$

$$2.) \quad \frac{\partial}{\partial t} \rho \langle v | + | \text{div } \rho \langle v | \rangle \langle v | = \rho \frac{D}{Dt} \langle v |$$

$$1.) \quad \frac{\partial}{\partial t} \rho \frac{v^2}{2} + \text{div } \rho \frac{v^2}{2} \langle v |$$

$$= \frac{v^2}{2} \left(\frac{\partial \rho}{\partial t} + \rho \text{div} \langle v | + \langle v | \nabla \rho \right) + \rho \left(\frac{\partial (v^2)}{\partial t} + \langle v | \nabla (v^2) \right)$$

$$\left\{ \begin{array}{l} \text{div} \langle v | \rho + \rho \text{div} \langle v | = \text{div} \rho \langle v | \end{array} \right.$$

$$\rightarrow = \frac{v^2}{2} \left(\frac{\partial \rho}{\partial t} + \cancel{\text{div} \rho \langle v |} \right) + \rho \left(\frac{\partial (v^2)}{\partial t} + \langle v | \nabla (v^2) \right)$$

$$- \text{div} \rho \langle v | + \text{div} \rho \langle v | = 0$$

$$= \rho \left(\frac{\partial (v^2)}{\partial t} + \langle v | \nabla (v^2) \right) = \rho \frac{D}{Dt} \left(\frac{v^2}{2} \right) \checkmark$$

$$2.) \quad \frac{\partial}{\partial t} \rho \langle v | + | \text{div} \rho \langle v | \rangle \langle v |$$

$$= \langle v | \frac{\partial \rho}{\partial t} + \rho \frac{\partial \langle v |}{\partial t} + \rho \langle v | \text{div} \langle v | + \langle v | \text{div} \rho \langle v |$$

$$= \langle v | \frac{\partial \rho}{\partial t} + \rho \frac{\partial \langle v |}{\partial t} + \rho \langle v | \text{div} \langle v | + \langle v | \nabla \rho + \rho \langle v | \nabla \langle v |$$

$$= \langle v | \left(\frac{\partial \rho}{\partial t} + \langle v | \nabla \rho + \rho \text{div} \langle v | \right) + \rho \left(\frac{\partial \langle v |}{\partial t} + \langle v | \nabla \langle v | \right)$$

$$\quad \quad \quad \text{div} \rho \langle v | \quad \text{div} \rho \langle v |$$

$$= \langle v | \left(-\text{div} \rho \langle v | + \cancel{\text{div} \rho \langle v |} \right) + \rho \left(\frac{\partial \langle v |}{\partial t} + \langle v | \nabla \langle v | \right)$$

$$\rho \left(\frac{\partial \langle v |}{\partial t} + \langle v | \nabla \langle v | \right) = \rho \frac{D \langle v |}{Dt} \checkmark$$

2- Using cartesian coordinates, prove for a fluid of constant density

$$\text{div } |\nabla v|^{\dagger} = 0$$

$$|\nabla v| = \begin{pmatrix} \frac{\partial v_x}{\partial x} & \frac{\partial v_x}{\partial y} & \frac{\partial v_x}{\partial z} \\ \frac{\partial v_y}{\partial x} & \frac{\partial v_y}{\partial y} & \frac{\partial v_y}{\partial z} \\ \frac{\partial v_z}{\partial x} & \frac{\partial v_z}{\partial y} & \frac{\partial v_z}{\partial z} \end{pmatrix} \rightarrow |\nabla v|^{\dagger} = \begin{pmatrix} \frac{\partial v_x}{\partial x} & \frac{\partial v_y}{\partial x} & \frac{\partial v_z}{\partial x} \\ \frac{\partial v_x}{\partial y} & \frac{\partial v_y}{\partial y} & \frac{\partial v_z}{\partial y} \\ \frac{\partial v_x}{\partial z} & \frac{\partial v_y}{\partial z} & \frac{\partial v_z}{\partial z} \end{pmatrix}, \text{div } |\nabla v|^{\dagger} = \begin{pmatrix} \frac{\partial}{\partial x} \left(\frac{\partial v_x}{\partial x} \right) + \frac{\partial}{\partial y} \left(\frac{\partial v_y}{\partial x} \right) + \frac{\partial}{\partial z} \left(\frac{\partial v_z}{\partial x} \right) \\ \frac{\partial}{\partial x} \left(\frac{\partial v_x}{\partial y} \right) + \frac{\partial}{\partial y} \left(\frac{\partial v_y}{\partial y} \right) + \frac{\partial}{\partial z} \left(\frac{\partial v_z}{\partial y} \right) \\ \frac{\partial}{\partial x} \left(\frac{\partial v_x}{\partial z} \right) + \frac{\partial}{\partial y} \left(\frac{\partial v_y}{\partial z} \right) + \frac{\partial}{\partial z} \left(\frac{\partial v_z}{\partial z} \right) \end{pmatrix}$$

$\frac{\partial}{\partial t} \rho$ $\nearrow$ 0, since constant density
 $\frac{\partial}{\partial t} \rho + \text{div } \rho v = 0 \Rightarrow \text{div } v = 0$

$$= \begin{pmatrix} \frac{\partial}{\partial x} \left(\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} \right) \\ \frac{\partial}{\partial y} \left(\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} \right) \\ \frac{\partial}{\partial z} \left(\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} \right) \end{pmatrix} = \nabla \text{div } v$$

plugging in $\text{div } v = 0$

$$= \nabla(0) = \underline{0}$$

$$\therefore \text{div } |\nabla v|^{\dagger} = 0$$